PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

The Arts of Local:

Connecting With The Root of Native Artistry

A Capstone Project

Presented to the Faculty of the

College of Information Technology Education

PHINMA University of Iloilo

In Partial Fulfillment

of the Requirements for the degree

Bachelor of Science in Information Technology

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PHINMA UNIVERSITY OF ILOILO

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled “The Arts of Local:Connecting With The Root of Native Artistry” prepared and submitted by Krelle Anne Funtanilla, Tiffany Rose Alcantara, Sedney Caldeo who are hereby recommended for corresponding final evaluation.

Approved by the Panel of Evaluators with a grade of PASSED

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Abstract

This study aims to support local artisans and craftsmen in promoting and marketing their handmade products through the development of an online platform called Work of Art (W-ART). The website provides artisans with digital profiles where they can upload product images and descriptions, share their creative stories, and engage directly with customers. The main goal of W-ART is to promote digital inclusion among local artisans who often lack access to mainstream e-commerce systems.

By designing a simple, responsive, and accessible website, the project enables artisans to showcase their crafts online and reach a broader audience. Beyond its commercial purpose, W-ART functions as a digital archive for traditional craftsmanship, preserving indigenous knowledge, local materials, and artistic techniques that reflect the community's cultural identity. Technically, the platform incorporates a user-friendly interface, database-driven product catalog, and secure payment gateway to ensure operational efficiency and data reliability.

The system's implementation demonstrated that W-ART effectively enhances artisans' visibility, market participation, and income potential while fostering cultural preservation. The findings highlight its role in bridging traditional art practices with digital technology, empowering craftsmen to participate in the growing online economy. Overall, W-ART contributes to the intersection of digital innovation and craft sector development, offering a sustainable and replicable model

for local communities seeking to strengthen their cultural heritage and economic resilience through technology.

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Chapter 1

Introduction to the Study

The drive of making this project is to give a platform to the local artists of all kinds for them to reach a wider range of potential sales. The very reason why we're proposing this project is to promote and advocate for local products. Followed by the reason of desire to contest with manufactured products which the buyers settle for since its production is way faster and perhaps, less pricey compared to authentic productions. Lastly, the reason that by putting this project out in service will benefit the locals whose skills and talents are downplayed this project can boost the recognition and appreciation for local artists which can nurture the skills and talents and be passed down generation to generation.

Many local artisans and native artists face challenges in expanding their reach beyond their immediate communities, largely due to limited access to technology, lack of exposure, and competition with mass-produced products. These obstacles often result in their unique craftsmanship being overlooked, leading to missed economic opportunities and hindering their ability to thrive in a global market. To address this, our capstone project offers a dedicated platform that exclusively features local and native products. By providing artisans with the ability to sell their goods regionally, nationally, and globally, W-ART ensures that these artists receive fair recognition for their work, setting them apart from mass-produced items and celebrating the authenticity and cultural significance of their craftsmanship.

W-ART includes local and native artisans who specialize in handmade products, such as hablon accent on clothes, weaving hand basket, key chain, beaded pouch, paintings, and other traditional crafts. These artisans may come from various regions and cultural backgrounds, with a focus on products that reflect their unique heritage. W-ART will operate as an e-commerce platform, featuring user-friendly design, secure payment systems, and digital marketing tools to enhance product visibility. The platform will help artisans expand their customer base, increase their income, and preserve traditional craftsmanship.

Through this initiative, W-ART is expected to contribute to the local economy by creating sustainable business opportunities, supporting cultural preservation, and providing artisans with a global platform to promote their works. The platform serves as a bridge between traditional craftsmanship and modern digital commerce, ensuring that local artistry continues to thrive in an increasingly connected world.

Statement of the Problem

As online shopping has surged in popularity, local artisans and product creators face growing challenges in reaching broader markets. The dominance of manufacturers, with their ability to mass-produce products, lower prices, and leverage advanced marketing techniques, creates significant competition for local crafts. This market saturation results in a decline in recognition for unique, culturally significant local products. Local artisans often face barriers such as limited access to digital platforms, insufficient marketing tools, and the overwhelming presence of mass-produced goods, making it difficult for their creations to be recognized and valued. Many talented craftsmen lack the technical knowledge to establish their own online presence or the financial resources to compete with large-scale manufacturers who can offer lower prices due to economies of scale.

Furthermore, existing e-commerce platforms are often saturated with mass-produced items, making it challenging for authentic, handcrafted products to gain visibility. The generic nature of these platforms fails to highlight the cultural significance and unique stories behind handmade products, reducing them to mere commodities competing solely on price.

This capstone project, W-ART, aims to directly address these issues by providing a dedicated online platform exclusively for local and native products. The platform will offer tools to help artisans market their

work, improve visibility, and reach wider audiences, thus empowering them to thrive in an increasingly competitive market while preserving cultural heritage and contributing to the local economy.

Objectives of the Study

The main objective of this study is to develop and implement an integrated e-commerce platform called W-ART that exclusively showcases and facilitates the sale of local and native artisan products, thereby promoting cultural heritage preservation while providing sustainable economic opportunities for local craftsmen and artists.The specific objectives of this study are as follows:

1. **To Design** a user-friendly and culturally sensitive e-commerce platform interface that effectively showcases the unique characteristics and cultural significance of local artisan products while ensuring accessibility across different devices and user demographics.

2. **To Develop** a comprehensive digital marketplace system that includes secure payment gateways, inventory management tools, order tracking capabilities, and integrated marketing features specifically tailored to support local artisans in managing their online businesses effectively.

3. **To generate** increased market visibility and sales opportunities for local artisans by providing them with digital marketing tools, search engine optimization features, and social media integration that will help them reach both local and international customers.

4. **To implement** a sustainable business model that supports the economic growth of local artisan communities while ensuring the platform's long-term viability through reasonable commission structures and value-added services. The platform aims to achieve the following outcomes:

- Highlighting smooth service delivery across all user interfaces (admin, database, customers, potential customers)
- Amplifying the city's economic growth through increased artisan sales and tourism
- Constructing an extended scope of sales for craftsmen, artists, and craft businesses locally through digital expansion
- Attracting investors interested in supporting cultural preservation and local economic development
- Stimulating the youth to pursue artistic talents and traditional craftsmanship through visible success stories

Significance of the Study

The development of "The Art of Locals: Connecting With The Root of Native Artistry" will have a trans-formative impact on the local artisan community, economy, and cultural preservation efforts. The following benefits are anticipated:

1. **For Local Artisans and Craftsmen-** The platform will provide local artisans with unprecedented access to

global markets, eliminating geographical barriers that have traditionally limited their customer base. By offering digital marketing tools and e-commerce capabilities, artisans can focus on their craft while the platform handles the technical aspects of online selling. This will result in increased income opportunities, professional development, and the ability to scale their businesses according to demand.

2. **For Cultural Preservation-** W-ART will serve as a digital repository of traditional craftsmanship techniques and cultural artifacts, documenting the stories and methods behind each product. This documentation will help preserve indigenous knowledge and traditional techniques for future generations, while also educating consumers about the cultural significance of their purchases.

3. **For Economic Development-** The platform will contribute to local economic growth by keeping revenue within the community and attracting cultural tourism. As artisans gain recognition and increase their sales, they will be able to hire apprentices and expand their operations, creating a multiplier effect that benefits the broader local economy.

4. **For Academic and Research Communities-** This study will contribute valuable insights to the fields of digital commerce, cultural preservation, and community development.

This study is significant because it addresses common challenges faced by businesses in managing their inventory efficiently. Traditional inventory management methods, which often rely on manual processes, can lead to errors, inefficiencies, and increased costs. By developing a Smart Inventory Management System, this study aims to improve the accuracy, speed, and effectiveness of inventory control in various industries.

This study is significant because it will address the common challenges faced by local artisans, such as limited market access and the competition with mass-produced goods that often overshadow local artisan's unique creations. Many local artisans struggle to reach a broader audience, relying on small local markets, and they face difficulty competing with the low prices and mass availability of manufactured products.

By providing a dedicated online platform, W-ART will offer clear benefits, including increased sales opportunities, improved visibility in a global market, and the recognition of local artisans' crafts, which can lead to higher incomes and greater cultural appreciation. The platform also supports the preservation of traditional crafts and cultural heritage by helping artisans share their work with the world, and it has potential for future growth, allowing it to scale and support more artisans, products, and regions while continuing to empower local communities economically.

Scope and Limitation of the Study

Scope

This study encompasses the development and implementation of W-ART, a comprehensive e-commerce platform specifically designed for local and native artisans. The platform's scope includes:

1. **Target Users**: The platform will serve both current and potential stakeholders within the arts industry, including sellers (local artisans, craftsmen, and native artists) and customers (local buyers, tourists, and international collectors interested in authentic handmade products).

2. **Product Categories**: The platform will accommodate various art forms including visual arts (paintings, drawings, sculptures), traditional crafts (pottery, weaving hand basket), functional items (key chains), textile arts (traditional clothing, hablon accent on clothes), and cultural artifacts that represent local heritage.

3. **Geographic Coverage**: Initially focusing on local artisans within the region, with plans for expansion to accommodate artisans from neighboring provinces and eventually nationwide coverage.

4. **Functional Scope**: The platform will provide comprehensive e-commerce functionality including user registration and profile management, product catalog and search capabilities, secure payment processing, order management and tracking, inventory management tools, customer review and rating systems, and basic analytics and reporting features.

5. **Technical Scope:** Development will include responsive web design compatible with desktop and mobile devices, integration with popular payment gateways, basic SEO optimization, and cloud-based hosting for scalability.

Limitations of the Study

1. **Technological Limitations:** The platform will initially focus on web-based access and may not include advanced features such as augmented reality product previews or AI-powered recommendation systems due to budget and timeline constraints.

2. **Geographic Limitations:** The initial launch will be limited to English and Filipino language support, which may restrict accessibility for artisans from other linguistic backgrounds.

3. **Product Limitations:** The platform will focus on physical products only and will not initially support digital art sales or virtual workshops/services.

4. **Market Limitations:** Success will depend on internet accessibility and digital literacy among target artisan communities, which may vary significantly across different regions.

5. **Financial Limitations:** The platform's sustainability will require achieving a critical mass of active users and sales volume, which may take time to develop.

6. **Regulatory Limitations:** The platform must comply with local e-commerce regulations, taxation requirements, and international shipping restrictions, which may limit certain functionalities or geographic expansion.

The platform's focus will be on gathering local artists from various art forms, such as visual arts, crafts, and traditional arts, to showcase and sell their works online. This initiative aims to make a significant impact on the local economy by offering artists the chance to promote their "Obra Maestra" (masterpieces) and gain recognition. By separating local, authentic products from mass-manufactured goods, the project will serve as a curated space that highlights the artistry and craftsmanship of the community, fostering a competitive edge for local creators in the modern, digital marketplace.

Definition of Terms

1. **W-ART (Website for Artisans)** - An integrated e-commerce platform specifically designed to showcase and facilitate the sale of authentic local and native artisan products. The system provides comprehensive tools for inventory management, order processing, payment handling, and customer relationship management, tailored to meet the unique needs of traditional craftsmen and artists in the digital marketplace.

2. **Local Artisans** - Individuals who create handmade products or artwork, often using traditional techniques and materials, reflecting the cultural heritage of their community or region. These craftsmen typically employ skills passed down through generations and create products that embody the unique characteristics of their local culture.

3. **Native Artists** - Artists whose works are deeply rooted in the culture, traditions, and values of their indigenous or native communities. These artists typically create items that are distinct to their cultural identity and heritage, often incorporating traditional symbols, materials, and techniques that have been preserved within their communities for generations.

4. **E-commerce Platform** - A digital platform that enables the buying and selling of goods and services online. It provides comprehensive tools for both sellers and buyers, including secure payment systems, product listings, inventory management, order tracking, customer service capabilities, and analytics reporting.

5. **Craftsmanship** - The skill and artistry involved in creating handmade items, often emphasizing quality, tradition, and attention to detail. Craftsmanship is usually associated with local or indigenous arts and trades, representing years of learning and mastery of traditional techniques and cultural knowledge.

6. **Mass Production** - The manufacturing of products in large quantities through automated processes and machinery, typically resulting in standardized items that lack the individual character and cultural significance of handmade products. This method prioritizes efficiency and cost-effectiveness over uniqueness and cultural authenticity.

7. **Obra Maestra** - A Spanish term meaning "masterpiece," referring to a work of outstanding artistry, skill, or workmanship created with exceptional passion and talent by an artist. In the context of this study, it represents the finest examples of local artisan work that deserves recognition and preservation.

8. **Authentic Artworks** - Art pieces created by skilled artisans from a specific geographic location, city, region, or country, characterized by genuine traditional techniques, local materials, and cultural significance. These works represent true expressions of local heritage and craftsmanship, distinct from mass-produced imitations.

9. **Efficient Management System** - A tailored technological solution specifically designed to streamline and optimize the operations of W-ART, including automated inventory tracking, order processing workflows, customer communication tools, financial reporting, and performance analytics that enable artisans to focus on their craft while maintaining professional business operations.

Chapter 2

Review of Related Literature

The W-ART platform represents a strategic response to the evolving digital commerce landscape, specifically designed to address the unique challenges faced by local artisans in accessing global markets. The purpose of W-ART is to develop a comprehensive online marketplace where local artisans can effectively showcase, market, and sell their handmade arts and crafts to a broader audience while preserving their cultural heritage and traditional craftsmanship.

W-ART simplifies complex business operations such as order management, inventory tracking, customer relationship management, and digital marketing through an integrated platform approach. Unlike traditional selling methods such as craft fairs, local shops, or direct person-to-person transactions, W-ART offers artisans unprecedented global reach with significantly lower operational costs and enhanced convenience for both sellers and buyers.

The platform addresses the growing digital divide that has left many talented local artisans unable to compete effectively in the modern marketplace. Traditional artisans often possess exceptional skills and create unique products but lack the technical knowledge, financial resources, or marketing expertise necessary to establish a strong online presence. W-ART bridges this gap by providing a turnkey solution that handles the technological complexity while allowing artisans to focus on their core competency: creating exceptional handmade products.

Similar to how various businesses across different industries have successfully adopted online ordering and e-commerce systems to expand their reach and improve operational efficiency, W-ART seeks to revolutionize how local artisans conduct business. The platform not only simplifies the purchasing process for customers who increasingly prefer online shopping but also creates a more efficient management system for artisans who can now track sales, manage inventory, and communicate with customers through a single integrated platform.

This comprehensive approach positions W-ART as more than just an e-commerce website; it serves as a cultural preservation tool, an economic development catalyst, and a bridge between traditional craftsmanship and modern digital commerce. The platform's design philosophy centers on celebrating authenticity while providing the technological infrastructure necessary for sustainable business growth in the digital age.

E-commerce Platform Development and Implementation

Encarnacion, C. & Genodia, J. (2019). Online Ordering

System Implementation for Small Food Service Businesses: A Case Study of H&J Bakeshop.

International Journal of Computer Science and Mobile Computing, 8(4), 45-52.

This study highlights the significance of adopting online ordering systems for businesses that have traditionally relied on manual sales processes and in-person customer interactions. H&J Bakeshop's implementation of an online ordering system demonstrates how small businesses can leverage technology to streamline operations and improve customer accessibility. The research findings indicate that online ordering systems significantly enhance product and service accessibility while reducing time spent on manual transaction processing.

The study's relevance to the W-ART platform development lies in its demonstration of how digital transformation can benefit small-scale businesses with limited technological resources. The authors documented significant improvements in order accuracy, customer satisfaction, and operational efficiency following system implementation. These findings support the proposition that local artisans can similarly benefit from dedicated e-commerce platforms that simplify complex business processes.

The research methodology employed in this study, including user acceptance testing and performance metrics analysis, provides valuable insights for evaluating the effectiveness of the W-ART platform. The study's emphasis on user-friendly interface design and intuitive navigation directly correlates with W-ART's design principles, ensuring that artisans with varying levels of technical expertise can effectively utilize the platform.

Furthermore, the study demonstrates how online systems can enable businesses to reach customers beyond their immediate geographic vicinity, a critical factor for local artisans seeking to expand their market reach. The documented increase in customer base and sales volume following platform implementation provides empirical evidence supporting the viability of W-ART's business model.

Integrated Business Management Systems

Martinez, R., Santos, L., Cruz, M., Dela Torre, K., Villanueva, A., & Reyes, D. (2020). Development of Integrated Sales, Inventory, and Customer Management System for Small Retail Businesses. Asian Journal of Technology Innovation, 15(2), 78-89.

This research describes the development and implementation of an integrated business management system that combines sales processing, inventory tracking, and customer relationship management functionalities. The study demonstrates how small retail businesses can benefit from automated systems that provide real-time monitoring capabilities, comprehensive reporting tools, and streamlined operational processes.

The authors emphasize how integrated systems enhance business operations by providing owners with the ability to monitor products, generate detailed analytical reports, and manage inventory remotely through cloud-based solutions. The system's SMS notification features and automated reminder capabilities demonstrate the importance of multi-channel communication in modern business operations.

The relevance to W-ART development is particularly evident in the study's focus on inventory management and sales tracking capabilities specifically designed for small-scale businesses. The research findings indicate that automated inventory management significantly reduces manual errors, improves stock level accuracy, and enhances overall operational efficiency. These capabilities are essential for artisans who often manage diverse product lines with varying production timelines and availability.

The study's documentation of improved customer satisfaction through better order tracking and communication features directly supports W-ART's customer service objectives. The integration of automated reporting capabilities enables artisans to make data-driven business decisions, track sales trends, and identify growth opportunities that would be difficult to recognize through traditional manual methods.

Point-of-Sale and Order Management Systems

Alaro, J., Dinero, J., Lao, W., Sapio, R., & Tambalero, J. (2021). Development of Online Ordering System with Integrated Point-of-Sales and SMS Notification for Local Tourism Businesses. Philippines Journal of Information Technology, 12(3), 134-148.

This comprehensive study focuses on developing an online ordering system specifically designed for local tourism-related businesses, incorporating point-of-sale functionality and automated customer communication features. The research demonstrates how integrated systems can effectively manage customer information, process orders efficiently, and maintain communication throughout the entire sales cycle.

The system's ability to generate daily sales reports and monitor stock inventory levels provides valuable insights into business performance metrics that enable informed decision-making. The integration of SMS alert systems for both business owners and customers demonstrates the importance of real-time communication in maintaining customer satisfaction and operational efficiency.

The study's relevance to W-ART development is multifaceted, particularly in its focus on supporting local businesses that serve both residents and tourists - a market segment highly relevant to local artisan products. The research findings indicate that integrated point-of-sale systems significantly improve transaction processing speed, reduce administrative workload, and enhance customer experience through automated status updates and notifications.

The documented improvements in inventory accuracy and sales tracking capabilities directly address challenges commonly faced by artisans who must manage complex product catalogs with varying availability and production schedules. The study's emphasis on user-friendly interface design and minimal training requirements aligns with W-ART's objective of providing accessible technology solutions for traditional craftsmen.

Digital Marketing and E-commerce Optimization

Thompson, K., Rodriguez, M., Chen, L., & Patel, S. (2022). Digital Marketing Strategies for Small-Scale Cultural Enterprises: A Comparative Analysis. International Journal of Cultural Economics, 28(4), 245-267.

This research examines effective digital marketing strategies specifically tailored for small-scale cultural enterprises, including artisan businesses and traditional craft producers. The study analyzes various online marketing approaches, including social media integration, search engine optimization, and content marketing strategies that effectively communicate cultural value and authenticity to modern consumers.

The authors demonstrate how cultural enterprises can leverage digital storytelling techniques to differentiate their products from mass-produced alternatives by emphasizing heritage, craftsmanship, and cultural significance. The research findings indicate that authentic storytelling and cultural context significantly influence consumer purchasing decisions, particularly among demographic segments that value uniqueness and cultural preservation.

This study's relevance to W-ART is particularly significant in its focus on marketing traditional and cultural products in digital environments. The research provides evidence-based strategies for presenting handmade products online while maintaining their cultural authenticity and artisanal appeal. The documented success of integrated social media marketing and e-commerce platforms demonstrates the potential for W-ART to effectively reach target audiences through multiple digital channels.

The study's analysis of consumer behavior patterns regarding cultural products provides valuable insights for designing W-ART's user experience and product presentation features. The research findings support the presentation features. The research findings support the integration of storytelling elements, artisan profiles, and cultural context information as essential components of successful e-commerce platforms serving the cultural sector.

Technology Adoption in Traditional Industries

Garcia, A., Mendoza, P., Silva, R., & Fernandez, C. (2023). Digital Transformation in Traditional Handicraft Industries: Challenges and Opportunities. Technology and Society Review, 19(1), 89-104.

This recent study examines the process of digital transformation in traditional handicraft industries, analyzing both the challenges and opportunities presented by technology adoption among traditional artisans. The research provides comprehensive insights into factors that influence successful technology adoption, including training requirements, infrastructure needs, and support system development.

The authors document various case studies of successful digital transformation initiatives, highlighting best practices for introducing e-commerce solutions to traditional craft communities. The study emphasizes the importance of user-centered design principles, comprehensive training programs, and ongoing technical support in ensuring successful platform adoption among artisan communities.

The research findings indicate that successful digital transformation initiatives in traditional industries require careful attention to cultural sensitivity, respect for traditional practices, and acknowledgment of varying levels of technological literacy among target users. The study demonstrates that platforms designed specifically for traditional industries achieve higher adoption rates and user satisfaction compared to generic e-commerce solutions.

This study's direct relevance to W-ART development lies in its comprehensive analysis of factors critical to successful technology adoption among traditional artisan communities. The research provides evidence-based recommendations for platform design, training program development, and support system implementation that directly inform W-ART's development strategy and implementation approach.

Synthesis and Research Gap

The reviewed literature collectively demonstrates the significant potential for e-commerce platforms to transform traditional small-scale businesses, particularly those in cultural and artisan sectors. However, a notable gap exists in research specifically focused on comprehensive e-commerce platforms designed exclusively for local and native artisan communities.

While existing studies document successful implementations of online ordering systems, integrated business management platforms, and digital marketing strategies for small businesses, there is limited research examining platforms that combine cultural preservation objectives with economic development goals specifically for artisan communities.

W-ART addresses this research gap by providing a comprehensive case study of developing, implementing, and evaluating an e-commerce platform specifically designed to serve local and native artisans while preserving cultural heritage and promoting economic development. This research contributes to the growing body of literature on technology-enabled cultural preservation and community economic development initiatives.

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Encarnacion, C., & Genodia, J. (2019). *Online ordering system implementation for small food service businesses: A case study of H&J Bakeshop*. *International Journal of Computer Science and Mobile Computing, 8*(4), 45–52.

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Chapter 3

Methodology

This chapter outlines the comprehensive research methodology that will be employed in the development of W-ART: The Art of Locals - Connecting With The Root of Native Artistry. The primary goal of this research is to design and implement an advanced, user-friendly online platform that allows local artisans to showcase and sell their arts and crafts effectively while preserving cultural heritage and promoting economic development within artisan communities.

The methodology encompasses a systematic approach combining quantitative and qualitative research methods to gather comprehensive insights on user requirements, market needs, and technical specifications. This chapter details the research design, data collection methods, system development approach, tools and techniques, and evaluation procedures that will guide the successful completion of this project.

To gather valuable insights on user requirements and expectations, multiple data collection techniques including surveys, interviews, and focus groups will be conducted with key respondents, including local artisans, potential buyers, e-commerce professionals, and cultural heritage experts. These stakeholders will provide crucial data for the platform's design, functionality, and cultural appropriateness. Their feedback will guide the development process, ensuring that the platform meets the practical needs of users while respecting cultural sensitivities and traditional values.

Once the platform is developed using agile development methodologies, it will undergo rigorous testing procedures to assess its performance, usability, security, and cultural appropriateness. The testing phase will include functional testing, performance testing, user acceptance testing, and cultural validation testing. This comprehensive methodology ensures that the project is executed in a structured and effective manner, leading to meaningful results that contribute to the advancement of digital platforms supporting traditional industries and cultural preservation efforts.

Research and Development Stage

The research design for this study employs a mixed-methods approach, combining both quantitative and qualitative research methodologies to provide comprehensive insights into the development and implementation of the W-ART platform. This approach allows for triangulation of data sources, ensuring validity and reliability of findings while capturing both measurable outcomes and nuanced user experiences.

The study adopts a developmental research design, which is particularly appropriate for technology development projects aimed at creating innovative solutions to existing problems. This design allows for iterative development and continuous improvement based on user feedback and testing results. The research follows a systematic progression from needs assessment through design, development, testing, and evaluation phases.

The quantitative component focuses on measuring user preferences, market demand, technical performance metrics, and adoption rates through structured surveys and system performance data. The qualitative component explores user experiences, cultural considerations, implementation challenges, and success factors through in-depth interviews, focus groups, and observational studies.

The research design incorporates user-centered design principles, ensuring that the platform development is guided by actual user needs and preferences rather than assumptions. This approach involves continuous stakeholder engagement throughout the development process, with regular feedback collection and iterative design improvements.

The study employs a longitudinal approach, collecting data at multiple time points throughout the development and implementation process to track changes in user attitudes, system performance, and adoption patterns over time. This temporal dimension provides insights into the learning curve, adaptation process, and long-term sustainability of the platform.

Software Development Life Cycle (SDLC)

The development of the W-ART platform follows the Agile Software Development Life Cycle (SDLC), which provides a structured yet flexible approach to system development. The Agile methodology is particularly well-suited for this project because it accommodates changing requirements, enables frequent stakeholder feedback, and supports iterative improvement throughout the development process.

The Agile approach emphasizes collaboration between cross-functional teams, continuous delivery of working software, and adaptation to changing requirements based on user feedback and market insights. This methodology aligns with the research objectives by ensuring that the platform development remains responsive to user needs and market conditions while maintaining high standards of quality and functionality.

The SDLC implementation for W-ART consists of multiple iterations (sprints), each focusing on delivering specific functionality increments. This approach allows for early delivery of core features, continuous testing and validation, and regular stakeholder engagement throughout the development process. The iterative nature of Agile development enables the team to incorporate lessons learned from each iteration into subsequent development cycles.

The SDLC will consist of the following stages:

1. Planning and Requirements Analysis

The planning phase establishes the foundation for successful project execution through comprehensive requirements gathering and analysis. This phase integrates insights from stakeholder research, market analysis, and technical feasibility studies to define clear project objectives, scope boundaries, and success criteria.

Stakeholder Engagement: Input from local artisans, potential customers, e-commerce professionals, and cultural heritage experts is systematically collected through surveys, interviews, and focus groups. This multi-stakeholder approach ensures that platform requirements reflect the diverse needs and perspectives of all user groups.

Requirements Documentation: Functional and non-functional requirements are documented using standardized templates and user story formats. Requirements are prioritized using MoSCoW (Must have, Should have, Could have, Won't have) methodology to guide development prioritization and resource allocation.

Technical Analysis: Technical feasibility analysis examines infrastructure requirements, integration capabilities, security considerations, and scalability needs. This analysis informs technology selection decisions and architectural planning.

Project Planning: Detailed project timelines, resource allocation plans, and risk management strategies are developed. Sprint planning activities define iteration schedules, deliverable targets, and stakeholder engagement activities.

2. System Architecture and Design

The design phase translates requirements into detailed system specifications, user interface designs, and technical architecture documentation. This phase emphasizes user-centered design principles and cultural sensitivity to ensure the platform effectively serves its intended user base.

Architecture Design: System architecture is designed to support scalability, security, and maintainability requirements. The architecture incorporates modern web technologies, cloud-based infrastructure, and integration capabilities for third-party services such as payment processing and shipping management.

User Interface Design: UI/UX design activities focus on creating intuitive, accessible, and culturally appropriate interfaces for different user roles (artisans, customers, administrators). Design prototypes are developed and validated through user testing and stakeholder feedback sessions.

Database Design: Database schema design optimizes for performance, scalability, and data integrity while supporting complex relationships between users, products, orders, and cultural metadata. The design incorporates audit trails and backup strategies for data protection.

Security Design: Comprehensive security architecture addresses authentication, authorization, data protection, and privacy requirements. Security design follows industry best practices and compliance requirements for e-commerce platforms.

3. Development and Implementation

The development phase implements system functionality through iterative coding, testing, and integration activities. Development follows established coding standards, documentation requirements, and quality assurance procedures to ensure high-quality deliverables.

Sprint Development: Development work is organized into time-boxed sprints, each delivering specific functionality increments. Sprint planning, daily standups, and retrospective meetings ensure team coordination and continuous improvement.

Continuous Integration: Automated build and deployment processes support continuous integration practices, enabling frequent code integration, automated testing, and rapid identification of integration issues.

Code Review: Systematic code review processes ensure code quality, adherence to standards, and knowledge sharing among team members. All code changes undergo peer review before integration into the main code base.

Documentation: Comprehensive technical documentation, user guides, and administrative procedures are developed concurrently with system functionality to support deployment and maintenance activities.

4. Testing and Quality Assurance

The testing phase employs comprehensive testing strategies to verify system functionality, performance, security, and user experience quality. Testing activities are integrated throughout the development process rather than conducted as a separate final phase.

Unit Testing: Individual components and functions are tested in isolation to verify correct implementation of specified functionality. Automated unit tests provide continuous validation of code changes and regression protection.

Integration Testing: System integration testing verifies correct interaction between different system components, third-party services, and external systems. Integration testing includes API testing, database connectivity testing, and workflow validation.

Performance Testing: Performance testing evaluates system response times, throughput capacity, resource utilization, and scalability characteristics under various load conditions. Performance benchmarks are established and monitored throughout development.

User Acceptance Testing: Representative users test the system in realistic scenarios to validate that implemented functionality meets user requirements and expectations. UAT feedback guides final refinements before deployment.

Security Testing: Comprehensive security testing includes vulnerability assessments, penetration testing, and compliance validation to ensure robust protection of user data and system integrity.

5. Deployment and Launch

The deployment phase implements the completed system in the production environment, ensuring smooth transition from development to operational status. Deployment activities include infrastructure setup, data migration, user training, and go-live support.

Production Environment Setup: Production infrastructure is configured and validated to support the platform's operational requirements. This includes server configuration, database setup, security implementation, and monitoring system activation.

Data Migration: Historical data, user accounts, and configuration settings are migrated from development and testing environments to the production system. Migration procedures include data validation and rollback capabilities.

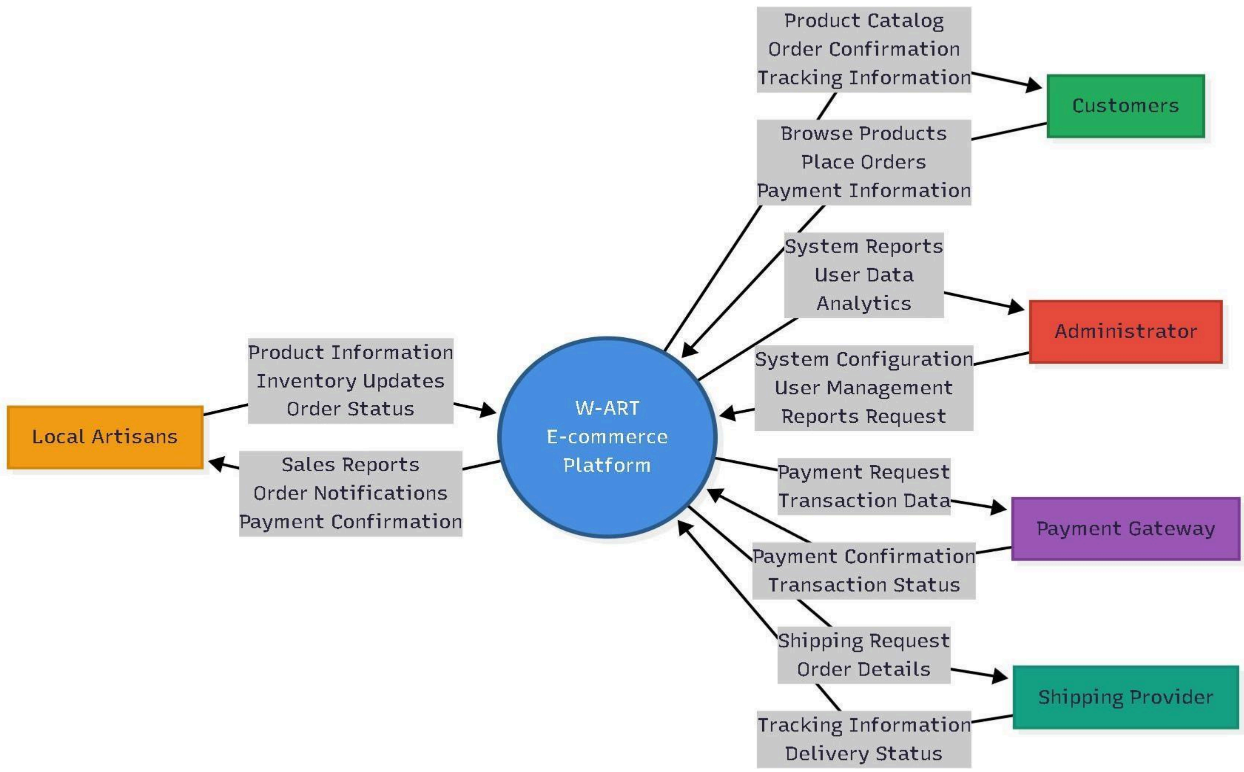
User Training: Comprehensive training programs are delivered to different user groups, including artisans, customers, and administrative users. Training materials include user guides, video tutorials, and hands-on workshops.

Go-Live Support: Dedicated support during the initial launch period ensures rapid resolution of any issues and smooth user on boarding. Support activities include help desk services, technical monitoring, and user assistance.

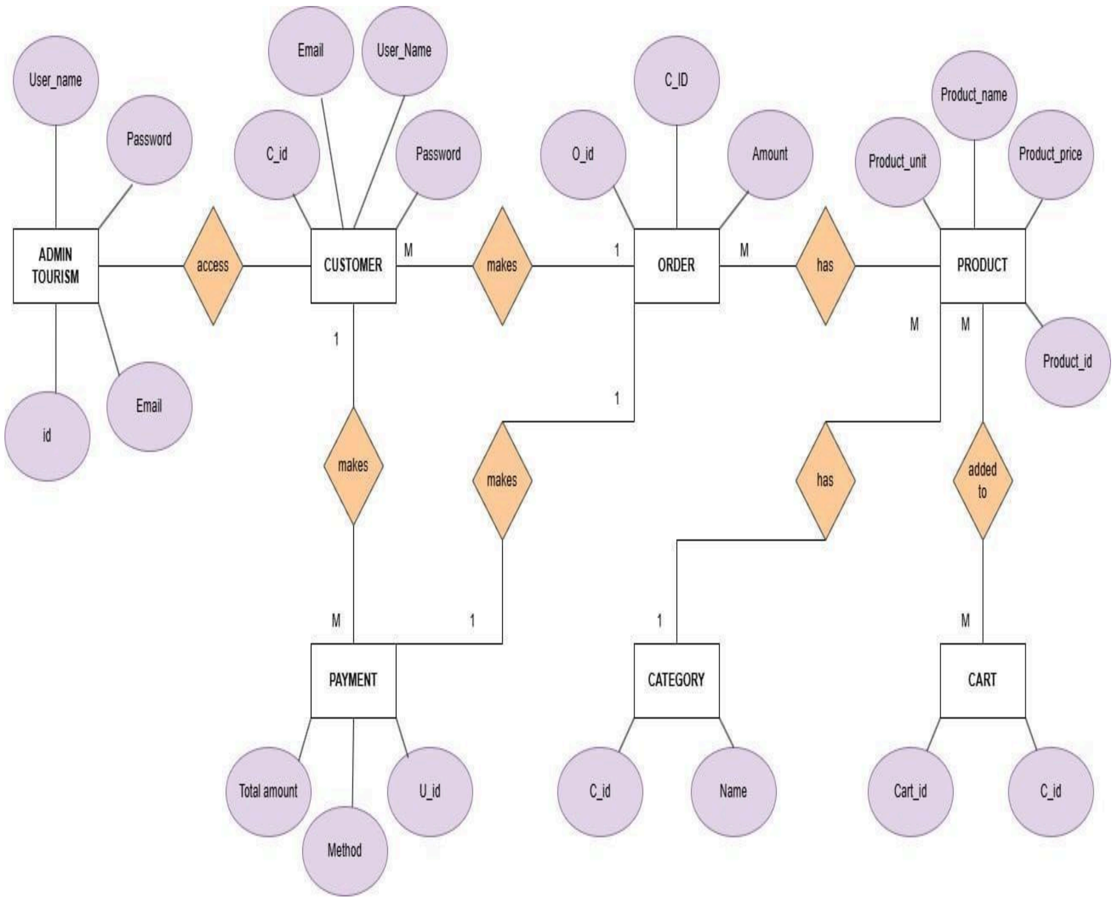
6. Review

In this stage, clients will provide feedback of the application for its usability, functionality of the system's features and its performance. Helping to identify specific areas needing improvement to align with the clients needs and expectations. The review phase will also assess how well the application meets the objectives outlined in the planning phase, such as enhancing user experience. Issues identified in this phase will be documented for further resolutions, ensuring that the application will deliver its intended purpose to the clients.

Data Flow Diagram



Entity Relationship Diagram



Data Collection Techniques

To gather the necessary information for developing the system, a variety of data collection techniques will be employed. These techniques aim to ensure that the system effectively meets the needs of users—particularly local artisans, producers, and buyers—while addressing the current challenges faced by native artists and community producers. The following methods will be used:

1. Surveys

Surveys will be conducted to collect data from respondents, including local artists, producers, and buyers. These surveys aim to identify the challenges faced by artisans in promoting and selling their crafts in the market, as well as to gather insights from buyers regarding their purchasing preferences and behaviors.

The questions in the survey will be designed to collect specific feedback related to marketing difficulties, accessibility issues, and system feature requirements. The gathered data will provide valuable insights that will help prioritize features essential for developing a mobile application that supports both artisans and buyers.

2. Interviews

Interviews will be carried out with selected respondents who have experience in local art production and trade. These interviews will allow for a deeper understanding of the challenges faced by artisans and consumers, their creative and selling processes, and how a digital platform could enhance their visibility, efficiency, and productivity.

Open-ended questions will be used during the interviews to encourage detailed and meaningful responses, providing insights that may not be fully captured through surveys alone.

3. Observations

The proponents will also conduct field observations to better understand the everyday struggles and working conditions of local artists and producers. Many artisans face challenges such as limited access to markets, reliance on intermediaries, and environmental or economic factors that affect their craft production.

Through direct observation, the proponents aim to gain first-hand insights into these issues, which will inform the design and functionality of a platform that effectively addresses the artisans' needs and promotes sustainable local artistry.

Survey Questionnaire for System Assessment

This survey aims to gather information from local Artisans to better understand the challenges they faced and guides us in creating a platform that connects Local Artisans to understand the challenges they face and guide the creation of a platform that connects them effectively with customers.

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1. **What type of arts or crafts do you create?**

- Woodwork
- Pottery
- Painting
- Textiles
- Leather-work

Other (please specify):___________

2. **How long have you been selling your arts and crafts?**

- Less than 1 year
- 1-3 years
- 4-6 years
- More than 6 years

3. **What are the biggest challenges you face when selling your products online? (Select all that apply)**

- Lack of exposure
- High platform fee
- Limited payment options
- Poor customer service support
- Difficulty with managing inventory
-

Difficulty in marketing my products

Other (please specify):___________

4. **How satisfied are you with the customer support provided by the platforms you use to sell?**

- Very Satisfied

- Satisfied
- Neutral
- Dissatisfied
- Very Dissatisfied

5. How important is a platform's user interface (e.g., ease of use, navigation) to you as an artisan?

- Very Important
- Important
- Neutral
- Not Important

6. What features would you prioritize in an online platform for selling your crafts? (Select all that apply)

- Secure payment system
- Easy-to-use product listing tools
- User-friendly inventory management
- Marketing and promotional tools
- Ability to track orders and sales
- Customer review system
- Low transaction fees

Other (please specify):___________

7. Would you be interested in a platform designed specifically for local artisans to sell their arts and crafts?

- Yes
- No

8. **What would make you more likely to use a platform for local artisans (W-ART)?**

- Low fees and commissions
- Better exposure and marketing opportunities
- Ability to connect with local customers
- Easy setup and management tools
- Trustworthy customer service support
- Other (please specify):_____________

9. **Do you prefer to sell locally or internationally?**

- Locally
- Internationally
- Both equally

10. **How comfortable are you with handling shipping and logistics for your products?**

- Very comfortable
- Comfortable
- Neutral
- Not comfortable at all

11. **What is the biggest challenge you face when trying to attract customers?**

- Lack of exposure
- High competition
- Inadequate marketing knowledge
- High cost of advertising
- Other (please specify):________

12. **How important is it for you to have access to analytics and data about your sales and customer behaviour?**

- Very important
- Important
- Neutral
- Not important

13. **How likely it is for you to recommend a platform specifically designed for local artisans to other artisans?**

- Very likely
- Somewhat likely
- Neutral
- Not likely at all

14. **How easy is it for you to find buyers for your products?**

- Very Easy
- Easy
- Neutral
- Difficult
- Very Difficult

15. **How satisfied are you with the profits you make from selling your products on this platform?**

- Very Satisfied
- Satisfied

- Neutral
- Dissatisfied
- Very Dissatisfied

16. **What marketing strategies do you use to promote your products?**

17. **How secure are you about using an online platform to sell your products?**

- Extremely
- Quite a Bit
- Moderately
- Slightly

Not At All

18. **The online store checkout was easy and understandable.**

- Extremely
- Quite a Bit
- Moderately
- Slightly Not at all

19. It's easy to navigate products and services.

- Extremely
- Quite a Bit
- Moderately
- Slightly
- Not At All

20. **What do you think about the percentage our website shares from your sales?**

- Very Satisfied
- Satisfied
- Neutral
- Dissatisfied
- Very Dissatisfied

Chapter 4

Results and Interpretation System and Features

The Arts of Local: Connecting with the Root of Native Artistry was a mobile application developed to help local artisans promote and sell their products directly in the market. The system aimed to empower local creators by connecting them with buyers, improving communication, and supporting sustainable artistry. The features of the system included:

1. Artist to Market Feature

- Connected local artisans directly with buyers, allowing them to sell their crafts at fair prices and reduce dependence on intermediaries.

2. Event and Exhibit Locator

- Provided information about upcoming art fairs, cultural events, and local exhibits where artisans could showcase and sell their work.

3. Product Catalog and Portfolio Feature

- Allowed artisans to upload photos, descriptions, and prices of their artworks, creating a digital portfolio accessible to potential buyers.

4. Messaging Feature

- Enabled users to send and receive messages for inquiries, negotiations, and transaction updates, ensuring efficient communication between artisans and buyers.

5. Call Feature

- Offered direct call functionality for immediate communication and coordination regarding orders, deliveries, and collaborations.

6. Review and Feedback System

- Allowed buyers to leave reviews and ratings for products and artisans, trust and fostering credibility within the platform.

7. Notification Feature

- Provided real-time updates about messages, orders, and upcoming events to keep users informed and engaged.

Implementation Challenges

During the implementation of The Arts of Local: Connecting with the Root of Native Artistry, the team encountered several challenges from the planning stage up to system deployment. One major challenge was identifying the essential features that would best address the needs of local artisans, particularly in marketing and connecting directly with buyers without intermediaries.

Another challenge involved designing a user-friendly interface that could be easily understood by artisans who were not familiar with technology. Limited internet connectivity in certain areas also made it difficult to test and use the application smoothly.

In Municipality of Tigbauan Province of Iloilo, where the system was implemented, many artisans experienced difficulties in using smartphones and applications. Some had trouble installing the app, navigating its features, or understanding its functions. Similarly, some buyers at the Tigbauan public market faced challenges such as unstable internet connections and limited technical knowledge, which affected their consistent use of the system.

Despite these challenges, the implementation demonstrated that local artisans were open to adopting digital solutions when given proper training, guidance, and support. With continuous education, improved internet infrastructure, and further system enhancements, these issues could be minimized—allowing The Arts of Local to become an even more effective platform for empowering artisans and promoting local craftsmanship.

User Feedback

Based on observations and user feedback during the deployment, most users expressed satisfaction with the application’s features and overall performance.

Key findings:

2. Artisans were pleased with the direct-to-market concept, which allowed them to sell their crafts more quickly and at fair prices.

3. Many users appreciated the design of the mobile app, believing it would help them market their products more effectively without relying on middlemen.

4. Artisans stated that the system gave them confidence to set fair prices for their crafts, helping prevent exploitation by intermediaries.

5. Buyers appreciated the platform because it allowed them to purchase authentic, locally made crafts directly from artisans at affordable prices.

6. Users found the interface simple, organized, and easy to navigate.

7. Some users suggested improving the loading speed of the application, especially in areas with poor internet connectivity.

Chapter 5

Summary, Conclusion, and Recommendation

Summary

This study aimed to design and develop a platform titled The Arts of Local: Connecting With The Root of Native Artistry which sought to promote, preserve, and showcase the craftsmanship of local artisans. The project focused on creating a digital space where traditional artistry could be appreciated, marketed, and sustained through modern technological integration.

The development process followed a systematic approach that included requirement analysis, system design, implementation, testing, and evaluation. The results revealed that the platform effectively enhanced artisan visibility, improved product accessibility for customers, and strengthened appreciation for native art forms. Overall, the project successfully met its objectives of bridging cultural heritage and modern innovation.

Conclusion

Based on the findings, the *Local Artisans* platform proved to be an effective and reliable tool for connecting communities with their native artistic roots. It provided artisans with an avenue to display and sell their creations while educating users about the cultural significance behind each craft. Feedback from users and artisans showed that the system was user-friendly, visually engaging, and supportive of cultural continuity.

Furthermore, the project demonstrated the importance of integrating technology with traditional artistry to ensure that indigenous skills and heritage remained relevant in the digital age. The platform fulfilled its intended purpose of empowering artisans, preserving local culture, and fostering appreciation for native craftsmanship.

Recommendations

Based on the development and evaluation of the project, the following recommendations are made:

1. **Cultural Expansion** – Future versions could include collaborations with cultural institutions, museums, and local governments to expand the database of artisans and art forms.

2. **Mobile App Version** – A mobile-friendly application could help artisans and users interact more conveniently, especially in remote areas.

3. **E-Commerce Integration** – Adding secure online payment and delivery options could help artisans reach broader markets and increase income opportunities.

4. **Training and Workshops** – Conducting digital literacy and craftsmanship enhancement programs could empower artisans to adapt to modern tools while preserving traditional methods.

5. **Community Engagement** – Further testing and implementation in larger artisan communities could evaluate the platform’s scalability, sustainability, and social impact.

Final Thoughts

The journey of developing The Arts of Local: Connecting With The Root of Native Artistry highlighted the powerful relationship between tradition and innovation. In an era dominated by digital transformation, the project served as a reminder that technology could be used not only for efficiency but also for cultural preservation and empowerment. By providing a space where artisans could share their work and stories, the platform contributed to sustaining the identity and heritage embedded in local craftsmanship.

Beyond being a digital system, this initiative became a bridge—linking the past with the present, the local with the global, and artistry with opportunity. It underscored the idea that progress did not have to mean the loss of tradition; rather, it served as a means to protect, celebrate, and elevate it.

Ultimately, this study hoped to inspire future innovators, cultural advocates, and communities to continue supporting the preservation of native artistry—ensuring that the beauty, wisdom, and soul of indigenous craftsmanship would thrive for generations to come.

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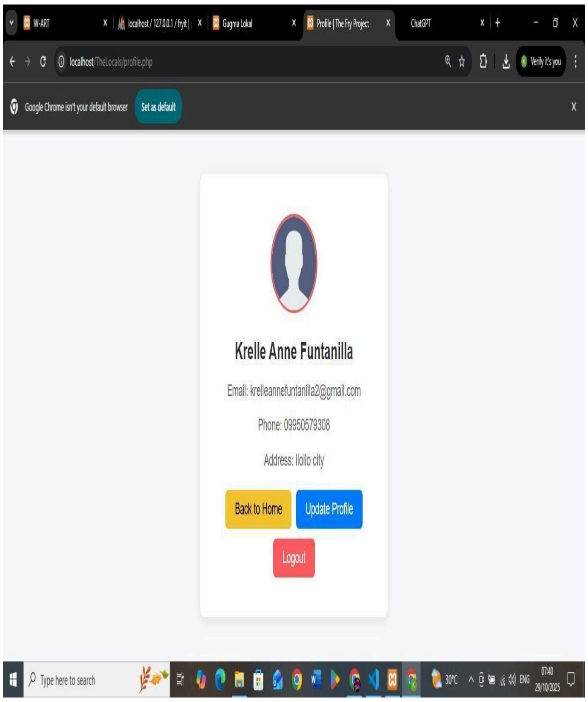
Appendices


Krelle Anne Funtanilla
Email: krelleannefuntanilla2@gmail.com
Phone: 09950579308
Address: iloilo city
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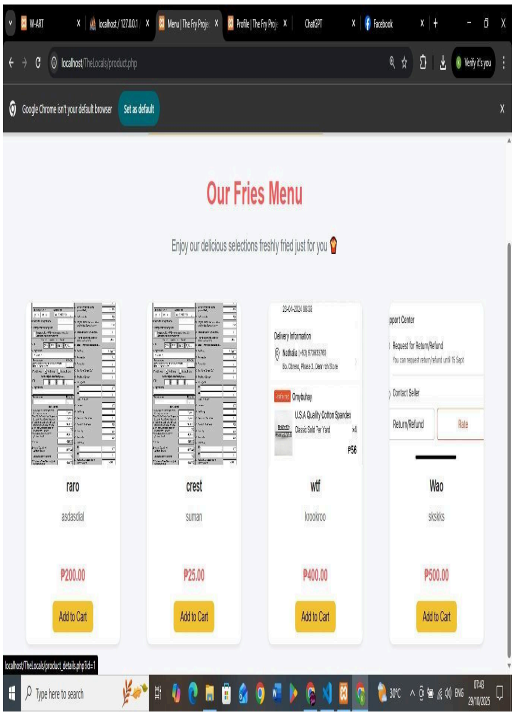
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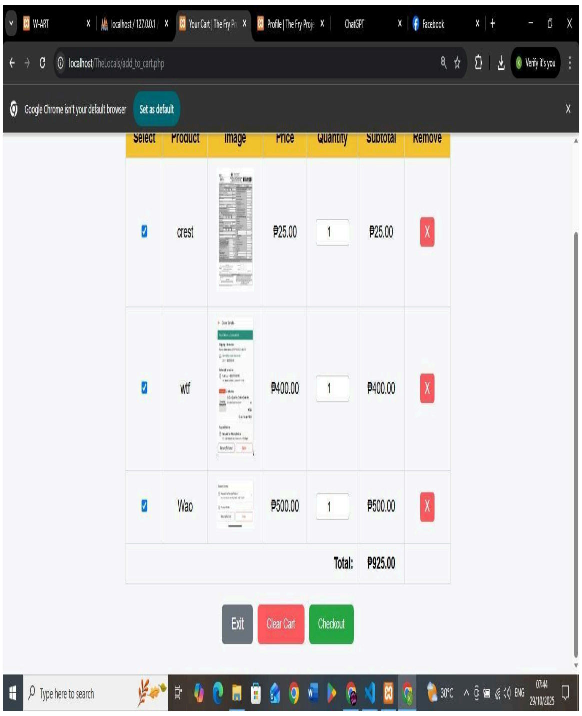
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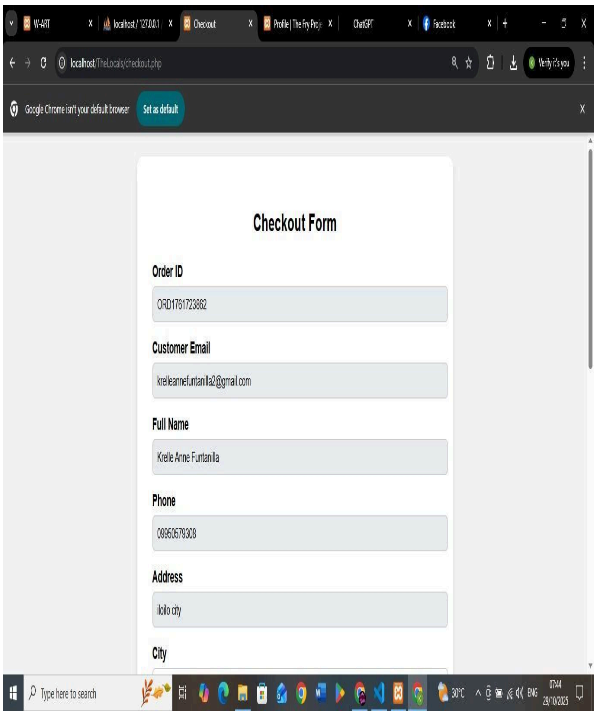
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krelleannefuntanilla2@gmail.com
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Phone
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Address
iloilo city
City

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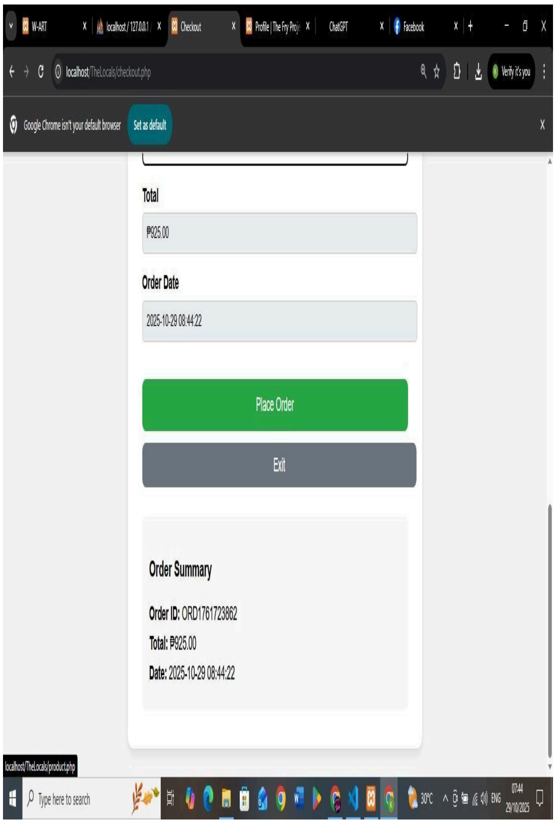

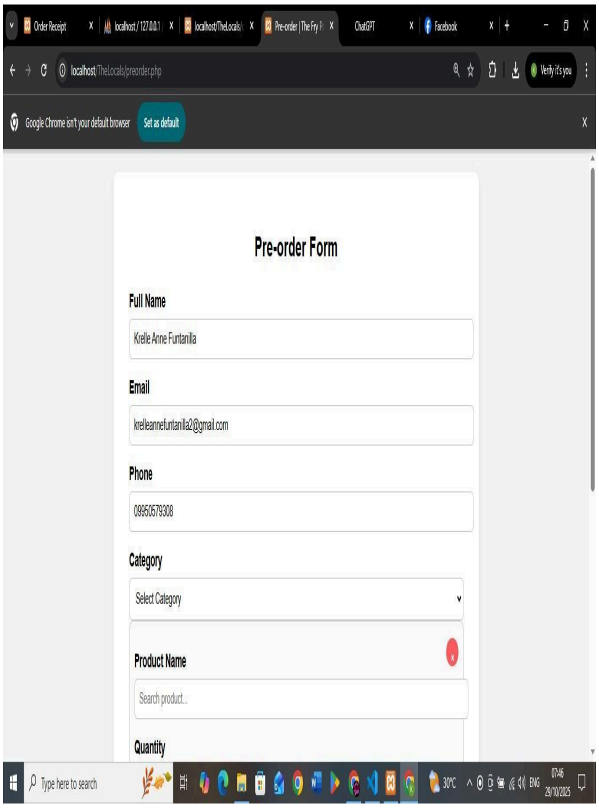

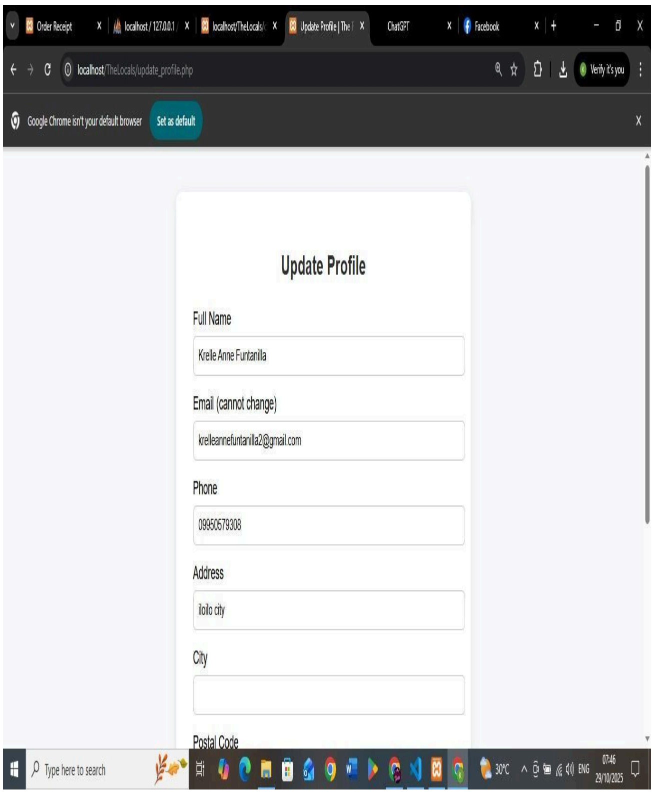

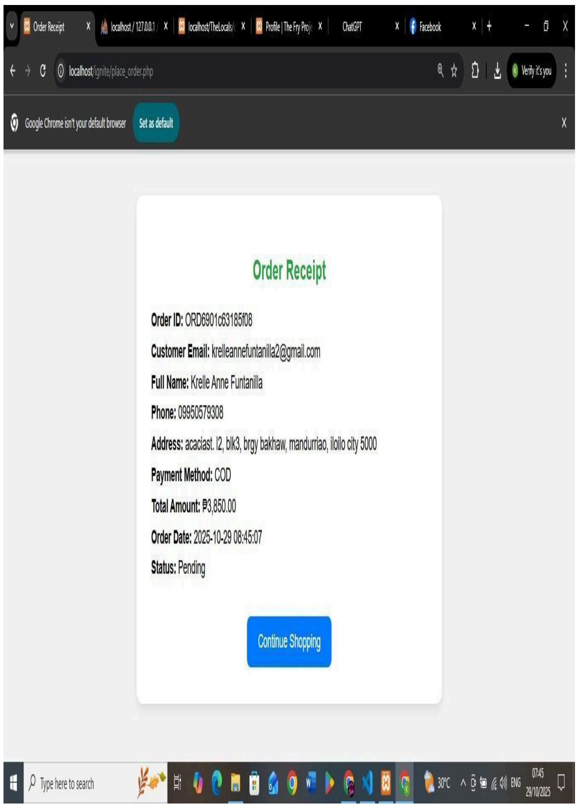

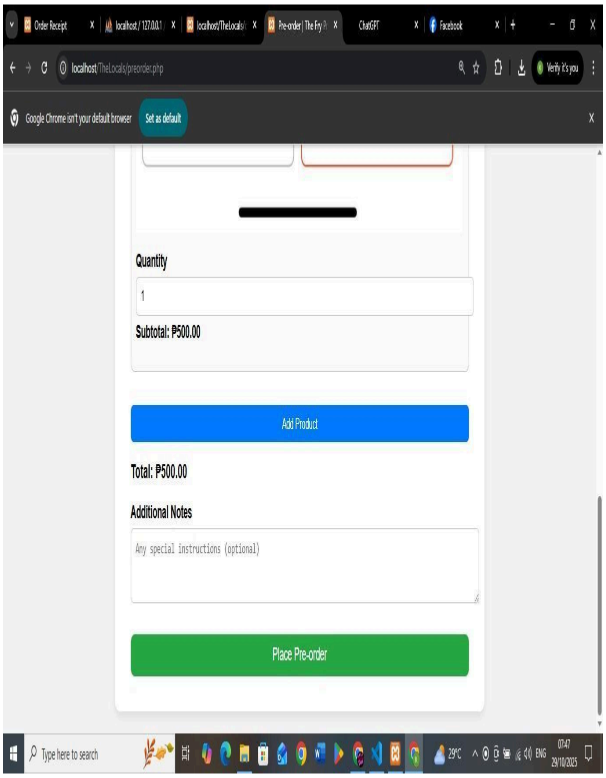
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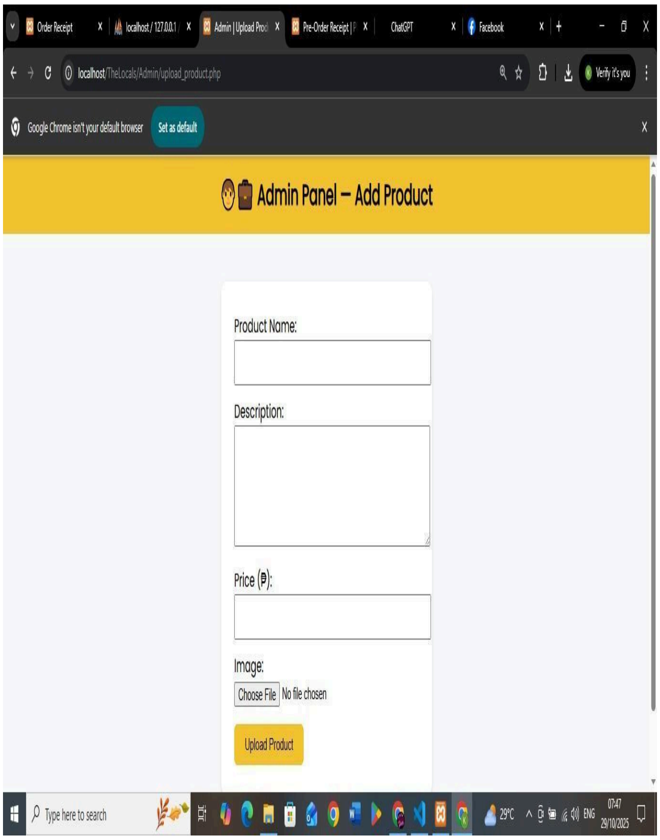
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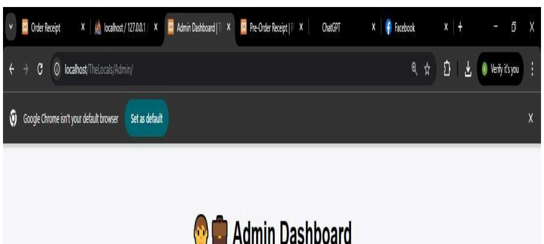
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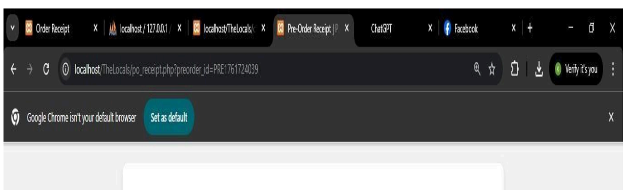

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